

BM4152 Biosignal Processing Assignment 3

- This is an individual assignment.
 - Use the data files (.npz) that are attached as instructed throughout the assignment.
 - Submission guidelines:
 - Report: Upload a PDF to Moodle. Include observations and discussions with relevant plots to support your answers.
 - Python scripts / notebooks: Upload a single ZIP file with all .py files (or a single .ipynb notebook) to Moodle. Name each script/section according to the question number. Include comments for readability.
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Introduction

This assignment consists of:

- Basic implementation of continuous and discrete wavelet transforms
- Use of Python scientific libraries for wavelet transforms
- Denoising and compression using wavelet techniques

1 Continuous Wavelet Transform

1.1 Introduction

The continuous wavelet transform is defined by

$$W(s, \tau) = \int x(t) \frac{1}{\sqrt{s}} \psi\left(\frac{t - \tau}{s}\right) dt,$$

where s is the scaling factor, τ is the translation, and ψ is the wavelet function. There are many wavelet families such as Haar, Shannon, Mexican hat, Morlet, and Daubechies that are useful for different applications (see Figure 1).

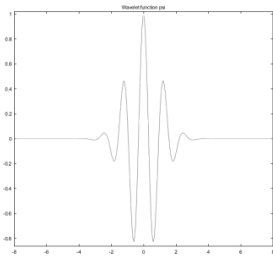


Figure 1.a. Morlet Wavelet

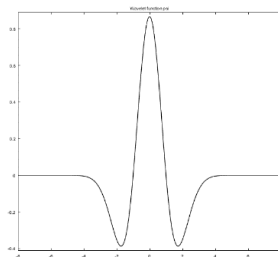


Figure 1.b. Mexican Hat Wavelet

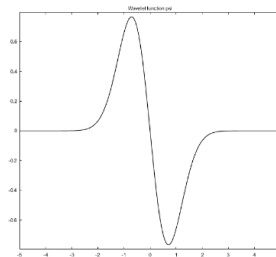


Figure 1.c. 1st derivative of Gaussian Wavelet

Figure 1: Wavelets

In this section, you will construct the Mexican-hat mother wavelet to check wavelet properties and then implement the CWT on a non-stationary signal.

1.2 Wavelet properties

- i. Given the Gaussian function $g(t)$, derive the Mexican-hat function $m(t)$:

$$m(t) = -\frac{d^2}{dt^2}g(t), \quad g(t) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2},$$

where $\mu = 0$ and $\sigma = 1$.

- ii. Calculate the normalizing factor of $m(t)$ such that

$$E = \int_{-\infty}^{\infty} m^2(t) dt = 1.$$

- iii. Hence, write the normalized Mexican-hat mother wavelet $\psi(t)$. Include the scaling factor s in the generic daughter-wavelet function.
- iv. Complete the provided script `wavelet_construction.py` to generate the Mexican-hat daughter wavelets for `s = 0.01 + 0.1*np.arange(20)`. Report the time-domain waveforms.
- v. Verify the wavelet properties of zero mean, unity energy and compact support (by observation) for each of the above daughter wavelets.
- vi. Using the same script, plot and comment on the spectra of daughter wavelets.

1.3 Continuous Wavelet Decomposition

- i. Create the following waveform in Python. Here n is a sample index and $f_s = 250$ Hz:

$$x[n] = \begin{cases} \sin(0.5\pi n/f_s), & 1 \leq n < \frac{3N}{2}, \\ \sin(1.5\pi n/f_s), & \frac{3N}{2} \leq n < 3N. \end{cases}$$

- ii. Apply the scaled Mexican-hat wavelets to $x[n]$. For each scale, achieve translations by convolving the signal with the constructed wavelet using `scipy.signal.fftconvolve(..., mode='same')`. Increase the scale resolution to `0.01*np.arange(1, 201)`.
- iii. Visualize the derived coefficients using `matplotlib.pyplot.pcolormesh(..., shading='auto')`. The result should look similar to Figure 2.
- iv. Comment on how the continuous-wavelet coefficients represent the time-varying frequency content of $x[n]$.

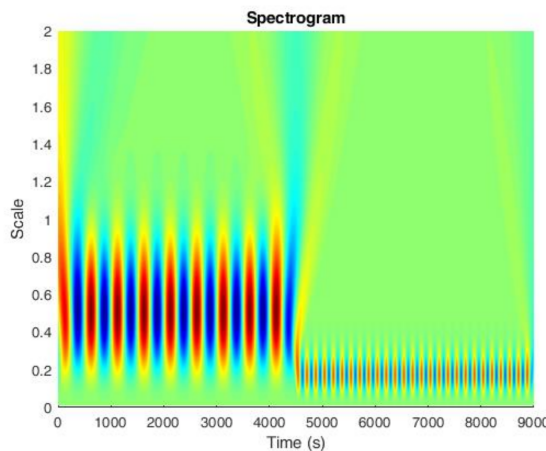


Figure 2: Spectrogram from the CWT

2 Discrete Wavelet Transform

2.1 Introduction

The drawbacks of CWT include highly redundant computations which leads to the requirement of additional computational power and time consumption. To avoid this, in the discrete wavelet transform (DWT), the scaling and translation are performed in a discrete manner.

For DWT, the equation for CWT is modified as follows.

$$\psi_{m,n}(t) = \frac{1}{\sqrt{s_0^m}} \psi\left(\frac{t - n\tau_0 s_0^m}{s_0^m}\right),$$

where s_0 is the scaling step size and τ_0 is the translation step size. Usually $s_0 = 2$ and $\tau_0 = 1$ are used for efficient analysis; m and n are the corresponding integer indices.

2.2 Applying DWT with PyWavelets

- i. Create the following waveforms in Python, where n is a sample index and $f_s = 512$ Hz:

$$x_1[n] = \begin{cases} 2 \sin(20\pi n/f_s) + \sin(80\pi n/f_s), & 0 \leq n < 512, \\ 0.5 \sin(40\pi n/f_s) + \sin(60\pi n/f_s), & 512 \leq n < 1024, \end{cases}$$

$$x_2[n] = \begin{cases} 1, & 0 \leq n < 64, \\ 2, & 192 \leq n < 256, \\ -1, & 256 \leq n < 512, \\ 3, & 512 \leq n < 704, \\ 1, & 704 \leq n < 960, \\ 0, & \text{otherwise.} \end{cases}$$

Corrupt the signals with AWGN at 10 dB SNR, call these signals y_1 and y_2 . Use `awgn(x, snr_db, rng)` helper function in `helper.py`. Plot the corresponding $x[n]$ and $y[n]$ on the same figure.

- ii. Observe the morphology of the wavelet and scaling functions of Haar and Daubechies tap 9 using PyWavelets' `wavefun()` method.
- iii. Calculate the 10-level wavelet decomposition of the signal using wavelet 'db9' and 'haar'.
- iv. Use the inverse DWT to reconstruct $A^{10}, D^{10}, D^9, \dots, D^2, D^1$ and verify that $y = A_L + \sum_{i=1}^L D_i$ by calculating the energy between original and reconstructed signal. Explain the steps followed.

2.3 Signal Denoising with DWT

- i. Plot the magnitude of wavelet coefficients (stem plot) of the above signal in descending order. Use `matplotlib.pyplot.stem`.
- ii. Select a threshold by observation assuming low magnitude coefficients contain noise. Reconstruct the signal with suppressed coefficients.
- iii. Calculate the root mean square error (RMSE) between the original and denoised signal. Plot the two signals on the same plot and interpret the results.
- iv. Repeat the same procedure with the Haar wavelet. Use the same stored noisy signal for a fair comparison.
- v. Compare the RMSE and the reconstructed wave morphology of the two waveforms with the two wavelets and comment on the suitability of the wavelets used.

2.4 Signal Compression with DWT

- i. You are given the aVR lead of an ECG sampled at 257 Hz. Obtain the discrete wavelet coefficients of the signal (use 'db9' and 'haar' wavelets).
- ii. Arrange the coefficients in the descending order and find the number of coefficients which are required to represent 99% of the energy of the signal
- iii. Compress the signal and find the compression ratio. Comment on the morphology of the reconstructed signal and the compression ratio.